

# Oluwatobi Melvyn Mayungbo

Data Analyst, Fraud and Risk

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## SUMMARY

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Data analyst focused on online fraud. Built transaction fraud detection with per-attack-type analysis and precision at fixed review capacity, a wallet and token risk tool with an explainable score, and a payment reconciliation system with human review. Works in SQL and Python and explains results to non-technical audiences.

## SKILLS

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**Payments and Risk:** Transaction Fraud Detection, Web3 Fraud Risk Scoring, Payment Reconciliation, Credit Risk Decisioning, Double-Entry Ledgers and Idempotency, Regulatory Compliance (CBN/NDPA), Demand Forecasting

**Data Analysis:** SQL, Pandas, Anomaly Detection, NumPy

**Engineering and MLOps:** Python, SQL, DuckDB, PostgreSQL, Redis, FastAPI, Docker, GCP Cloud Run, AWS ECS Fargate, Railway, Prefect, dbt-core, MLflow, Evidently, CI/CD (GitHub Actions), Git, Pytest, Streamlit, React

**Machine Learning:** PyTorch, Scikit-learn, LightGBM, XGBoost, Deep Learning, Sequence Models (GRU, TCN, Transformers), Model Calibration (Isotonic Regression, Temperature Scaling), SHAP, Feature Engineering

**Statistics and Causal Inference:** Statistical Inference, Experiment Design, Power Analysis, CUPED, Causal Inference (S/T/X-Learners, Causal Forests), Uplift Modeling, Qini and AUUC, Time-Series Backtesting, Logistic Regression, WOE Scorecards

## PROJECTS

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### Sequence-Based Transaction Fraud Detection System | [GitHub](#)

*Tech: Python, PyTorch, LightGBM, ONNX Runtime, FastAPI, Docker, SHAP*

- Benchmarked all architectures against a strong LightGBM baseline on engineered velocity and deviation features, reporting PR-AUC and precision at fixed review capacity (top 0.1%, 0.5%, 1%) across 3 seeds, with a per-attack-type breakdown covering card testing, account takeover and merchant compromise; sequence models cut expected cost per transaction 27% under an explicit cost model.
- Built and compared three deep sequence architectures (GRU, Temporal Convolutional Network and a causally masked Transformer encoder) over customer transaction histories, with learned embeddings for high-cardinality merchant (~2.9k vocabulary), MCC and device-type categoricals; all models under 115k parameters and CPU-trainable in under 6 minutes.
- Handled extreme class imbalance with focal loss and weighted sampling and isolated their failure modes: focal loss degraded probability calibration (ECE 0.0063) while weighted sampling cost 0.042 PR-AUC. Corrected calibration with temperature scaling and isotonic regression, reducing ECE to 0.0005.

### web3-risk-mcp: Web3 Fraud Risk MCP Server | [GitHub](#)

*Tech: Python, Model Context Protocol (MCP), Etherscan V2, GoPlus, DexScreener, Docker, GitHub Actions*

- Built an MCP server that gives AI assistants (Claude Desktop, Cursor) six read-only tools to check a crypto wallet, token or smart contract for risk before anyone interacts with it, returning a 0 to 100 risk score with a reason for every point, across Ethereum, Base, Arbitrum One, Polygon PoS and BNB Chain.
- Detects honeypot tokens, rug pull signs, hidden owner powers (mint, blacklist, pause), buy and sell tax, holder concentration, unlocked liquidity, and fund links to mixers, sanctioned wallets, exploiters and phishing addresses over 1 or 2 hops.

### Reckon: Payment Reconciliation and Review System | [GitHub](#) | [Live App](#)

*Tech: Python, PyTorch, DuckDB, FastAPI, React, Docker, Railway, PostgreSQL*

- Built and deployed a payment reconciliation system processing multi-source transaction feeds to detect uncollected revenue, duplicate settlements and payout discrepancies, combining deterministic matching rules with ML-assisted discrepancy resolution.
- Designed a human-in-the-loop review interface in React showing match confidence scores, audit trails and suggested resolution actions to streamline exception handling.

## EXPERIENCE

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**Machine Learning Instructor, SQI College of ICT**

March 2026 to Present

- Teach machine learning and deep learning to student cohorts, covering statistical foundations, classical ML algorithms, neural architectures, sequence models and production evaluation metrics.
- Instruct students on data analysis, feature engineering and model building with PyTorch, Scikit-Learn, Pandas and NumPy.

**Internal Auditor, NISER**

May 2024 to July 2024

- Analysed financial and operational records to identify inconsistencies and irregularities, performing manual anomaly detection across complex transactional datasets.

**EDUCATION**

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**B.Sc. Statistics, University of Ibadan**

January 2021 to March 2025

**Professional Diploma in Artificial Intelligence, SQI College of ICT**

July 2025 to Present

**CERTIFICATIONS**

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- Google Data Analytics Professional Certificate
- Google Advanced Data Analytics Professional Certificate
- Claude Code In Action